

Kreeda-Prerana Scout

(Grassroots Sports Talent Tracker)

Project Requirements Document

Android App Development using Generative AI | Sports Analytics Domain

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Domain	Sports Analytics / Rural Education	Platform	Android (Java / Room DB)
Status	Final — For Academic Review	Prepared For	Academic Project Evaluation

1. Problem Statement

Talented young athletes in rural schools across India often go unnoticed because there is no structured way to record their physical milestones — sprint times, jump heights, throw distances. Physical education teachers in these schools rely on memory or scattered paper records, making it impossible to track improvement over months or identify students who are ready for district-level competitions. Without a reliable, offline-capable mobile tool, promising sportspersons in remote areas miss critical opportunities for recognition and advancement.

2. Product Vision & Detailed Description

Kreeda-Prerana Scout is a **Grassroots Sports Talent Tracker** designed exclusively for rural school physical education teachers. It replaces scattered paper registers with a fast, intuitive mobile talent tracker. The core design principle is **speed and simplicity** — recording any trial must take no more than a few taps. At the end of each practice session, the teacher instantly sees **Leaderboard Rankings** and **District Level Ready** badges, empowering even the most resource-constrained schools with actionable sports analytics.

3. App Usage & User Flow

- **Athlete Profile Management:** A streamlined workflow: Tap FAB → Fill Name, Age, Class, School, Sport → Save. Supports 10 sport categories including Athletics, Kabaddi, Kho-Kho, Wrestling, Boxing, Archery, Hockey, Football, Volleyball, and Badminton. Profiles are displayed in a RecyclerView with instant search.
- **Trial Logger (Stopwatch):** High-precision stopwatch using `SystemClock.elapsedRealtime()` with 10ms refresh rate displaying MM:SS.HH format. Two input modes: Stopwatch mode for timed events (sprints, runs) and Manual Entry mode for distance events (long jump, shot put). Each trial is saved with automatic benchmark checking.
- **Batch Entry:** Record an entire class of 30+ students at once. All registered athletes are displayed with inline EditText fields for value entry. One-tap 'Save All' triggers `bulk insertAll()` to Room DB with automatic District Level Ready badge checks for every entry.
- **Leaderboard & Milestone Badges:** Rankings sorted per sport event — MIN (best time) for timed events, MAX (best distance) for distance events. Gold, silver, and bronze visual indicators for top 3. Automatic "District Level Ready" badge awarded when performance crosses 13 predefined sport benchmarks (e.g., 100m < 14 seconds).

- **Talent Curve Graph:** MPAndroidChart LineChart showing a student's performance improvement over time. Cubic Bezier smoothing, filled area below the curve, benchmark LimitLine, date-based x-axis labels, and improvement percentage calculation. Gives teachers a clear visual of each athlete's growth trajectory.

4. Technical Implementation

Component	Technology / Approach	Notes
Database	Room DB (SQLite abstraction)	Persistent local storage for all athletes and trials; works fully offline with no internet required
Entities	Athlete + TrialRecord with @ForeignKey	CASCADE delete ensures data integrity; @Index on athleteId for query performance
Business Logic	Executors + runOnUiThread()	Background thread DB operations prevent UI blocking; results posted to main thread
Timer	SystemClock.elapsedRealtime() + Handler	10ms Runnable callback; monotonic clock unaffected by system time changes; 2-decimal precision
User Interface	Material Design Components + XML	MaterialCardView, RecyclerView, Spinner, FAB; consistent Material Components theme throughout
Charting	MPAndroidChart v3.1.0 LineChart	Cubic Bezier mode, LimitLine for benchmarks, IndexAxisValueFormatter for dates, 800ms animation
Badge System	BenchmarkUtils utility class	13 predefined thresholds in HashMap; meetsDistrictBenchmark() auto-awards badges on save
Batch Processing	RecyclerView + insertAll()	Inline EditText per athlete; adapter array stores values; bulk DAO insert for 30+ students

5. Impact Goals

Grassroots Talent Discovery	Systematically identify and track promising young athletes in rural schools who would otherwise go unnoticed, bridging the gap between village playgrounds and district-level competitions.
Data-Driven PE Teaching	Empower physical education teachers with quantitative performance data, Talent Curve graphs, and automatic benchmarks — transforming subjective assessments into objective, evidence-based talent identification.
Rural Sports Infrastructure	Contribute to India's grassroots sports ecosystem by providing a free, offline-first tool that works in areas with no internet connectivity — aligning with Khelo India's vision of nurturing talent at the school level.

6. Success Criteria for Students

ID	Criterion	Acceptance Condition
SC-01	Timer Precision	The stopwatch must be accurate to 2 decimal places (centiseconds) using SystemClock.elapsedRealtime() with a 10ms Handler refresh rate displaying MM:SS.HH format.
SC-02	Batch Entry Capacity	The Batch Entry screen must handle 30+ students at once with inline EditText per athlete, and save all records in a single bulk operation via insertAll().

SC-03	Milestone Badge Accuracy	The app must automatically award a "District Level Ready" badge when an athlete's performance crosses any of the 13 predefined sport benchmarks (e.g., 100m Sprint < 14.0 seconds).
SC-04	Leaderboard Correctness	The Leaderboard must use MIN aggregate for timed events (lower = better) and MAX aggregate for distance events (higher = better), with gold/silver/bronze indicators for top 3 positions.
SC-05	Talent Curve Clarity	The MPAndroidChart Talent Curve graph must display a clear line chart with cubic Bezier smoothing, a benchmark LimitLine, date labels on x-axis, and an improvement percentage calculation.
SC-06	Offline Operation	The app must function completely offline using Room Database for all data persistence — no internet connection required for any core feature.
SC-07	Profile Management	Athletes must be creatable via a dialog form with Name, Age, Class, School, and Primary Sport (10 options). Profiles must be viewable, and deletable with CASCADE removal of all associated trial records.